

Abstract

This thesis comprises of three results each of which dealt in separate chapters. First chapter is of introductory nature, as the title suggest. And the other three chapters are devoted to three different problems. Following is a brief introduction to our results.

1. Let G be any finite abelian group of rank r with invariants $n_1, n_2, \dots, n_r$. In other words, $G = \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \dots \oplus \mathbb{Z}_{n_r}$ where n_i 's are integers satisfying $1 < n_1 | n_2 | \dots | n_r$. The *Davenport constant* of a group G is defined as the smallest positive integer t such that every sequence of length t of elements of G has a non-empty zero-sum subsequence. It has been conjectured by Śliwa that, $D(G) \leq \sum_{i=1}^r n_i$. Thinking in the direction of this conjecture we have obtained the following upper bound on Davenport constant $D(G)$, of G ,

$$D(G) \leq n_r + n_{r-1} + (c(3) - 1)n_{r-2} + (c(4) - 1)n_{r-3} + \dots + (c(r) - 1)n_1 + 1,$$

where $c(i)$'s are Alon-Dubiner constants [10] for respective i 's. Also we shall give an application of Davenport's constant to Quadratic sieve.

2. Let G be a finite abelian group with $\exp(G) = e$. Let $s(G)$ (respectively, $\eta(G)$) be the minimal positive integer t with the property that any sequence S of length t of elements of G contains an e -term subsequence (respectively, a non-empty subsequence of length at most e) of S with sum zero. For the group of rank at most two this constant has been determined completely (see [45]). Looking at the problem for groups of rank greater than 2 gave rise to this result. Our problem is to determine value of $s(C_{nm}^r)$ under some constraints on n, m , and r .

Let n, m and r be positive integers and $m \geq 3$. Furthermore, $\eta(C_m^r) = a_r(m - 1) + 1$, for some constant a_r depending on r and n is a fixed

integer greater than or equal to,

$$\frac{m^r(c(r)m - a_r(m - r) + m - 3)(m - 1) - (m + 1) + (m + 1)(a_r + 1)}{m(m + 1)(a_r + 1)}$$

and $s(C_n^r) = (a_r + 1)(n - 1) + 1$. In the above lower bound on n , $c(r)$ is the Alon-Dubiner constant. Then $s(C_{nm}^r) = (a_r + 1)(nm - 1) + 1$.

3. Given an abelian group G of order n , and a finite non-empty subset A of integers, the *Davenport constant of G with weight A* , denoted by $D_A(G)$, is defined to be the least positive integer t such that every sequence $(x_1, \dots, x_t)$ with $x_i \in G$ has a non-empty subsequence $(x_{j_1}, \dots, x_{j_l})$ and $a_i \in A$ such that $\sum_{i=1}^l a_i x_{j_i} = 0$. Similarly, $E_A(G)$ is defined to be the least positive integer t such that every sequence $(x_1, \dots, x_t)$ of length t of elements of G has a subsequence $(x_{j_1}, \dots, x_{j_n})$ such that $\sum_{i=1}^n a_i x_{j_i} = 0$, for some $a_i \in A$. When G is of order n , one considers A to be a non-empty subset of $\{1, \dots, n - 1\}$. If G is the cyclic group $\mathbb{Z}/n\mathbb{Z}$ we denote $E_A(G)$ and $D_A(G)$ by $E_A(n)$ and $D_A(n)$ respectively.

Here we extend some results in an article of Adhikari et al. [5] and determine bounds for $D_{R_n}(n)$ and $E_{R_n}(n)$, where $R_n = \{x^2 : x \in (\mathbb{Z}/n\mathbb{Z})^*\}$ and $(\mathbb{Z}/n\mathbb{Z})^*$ is a group of units modulo n . We follow some line of arguments in [5] and use a recent result of Yuan and Zeng [79], a theorem due to I. Chowla [24] and Kneser's theorem [52].